



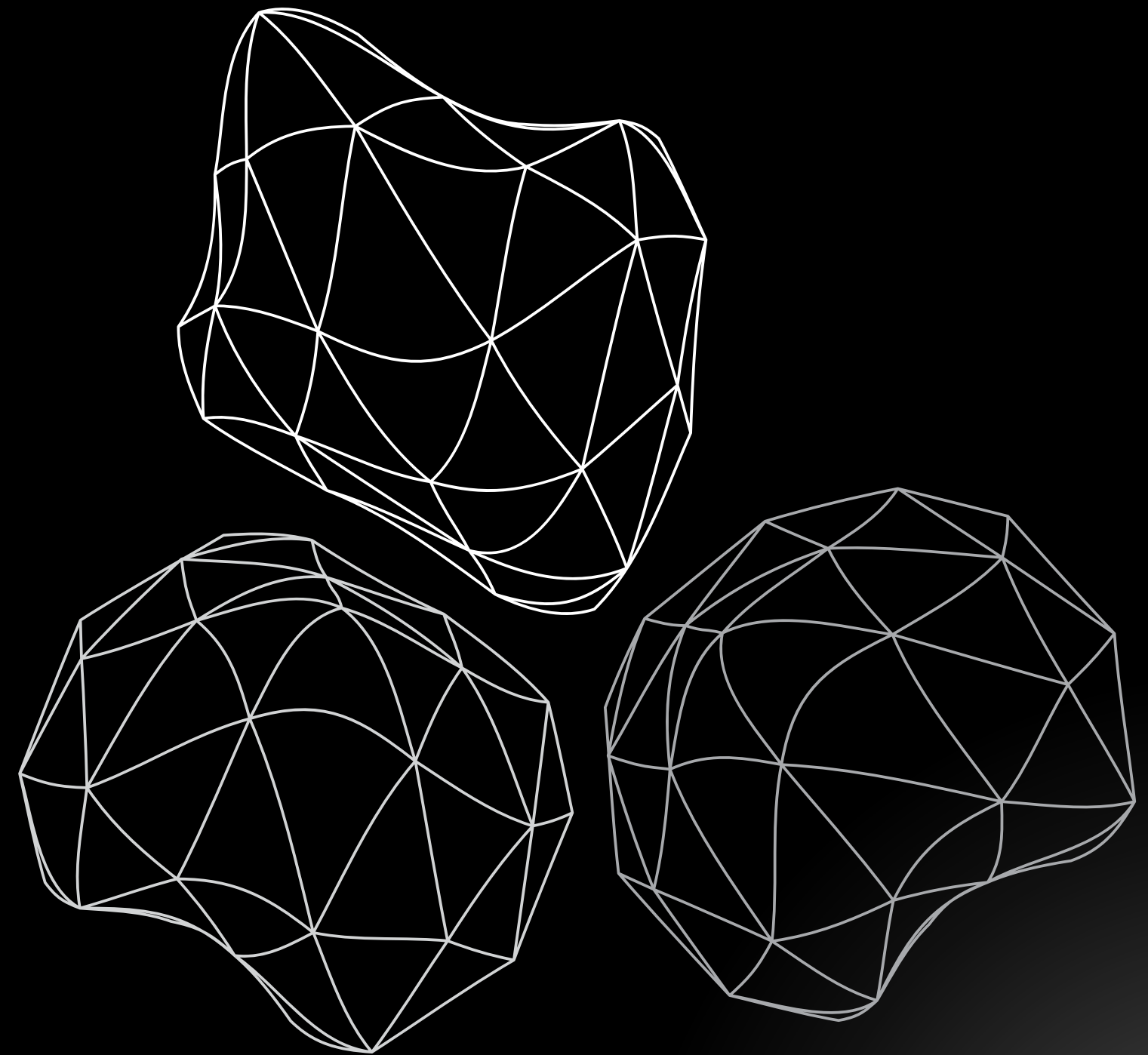
GSAPI

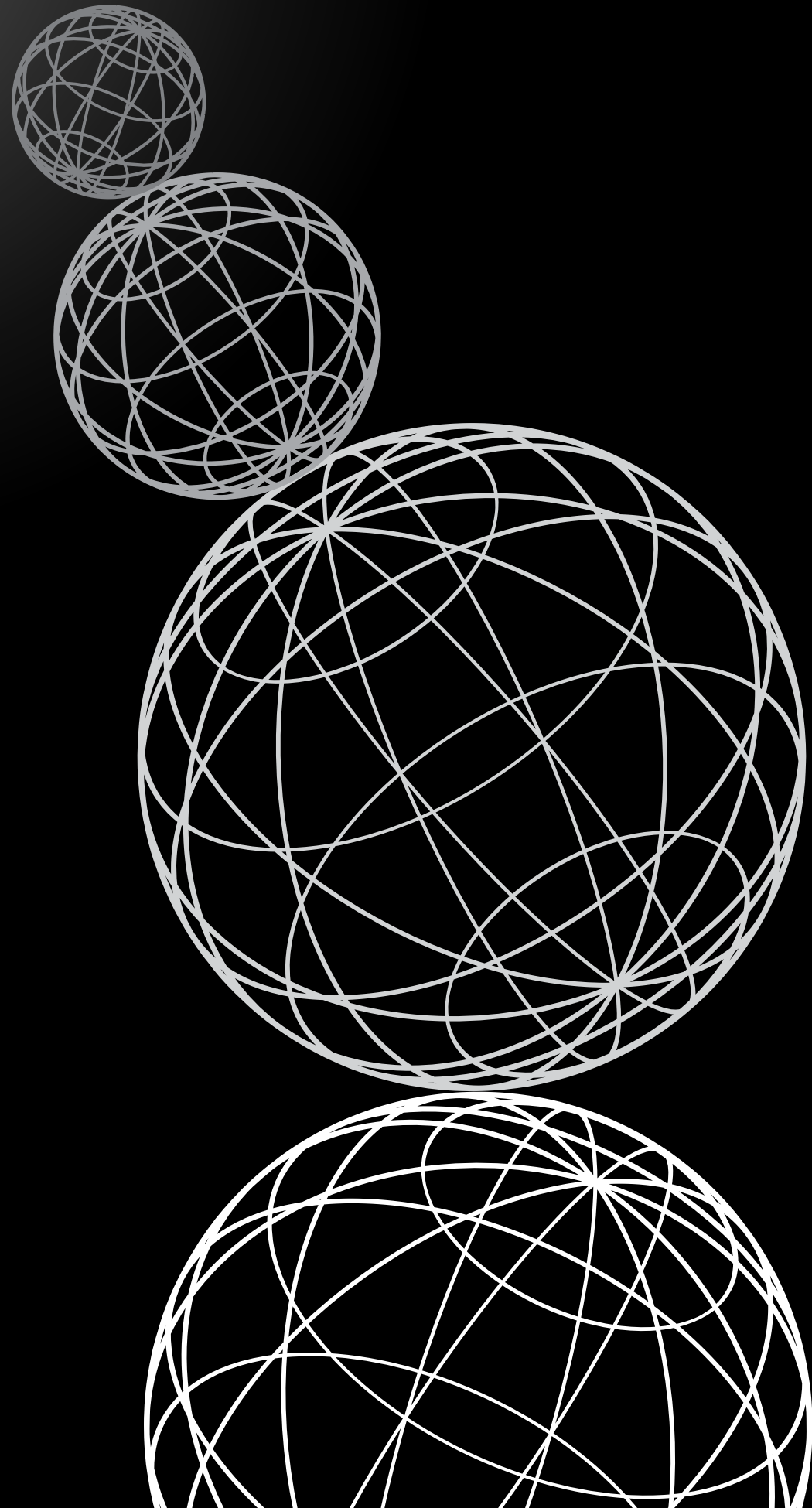
PRESENTATION

GAMESTATISTICS API

Und wie ich eure Daten rette

MARIA LINDLING

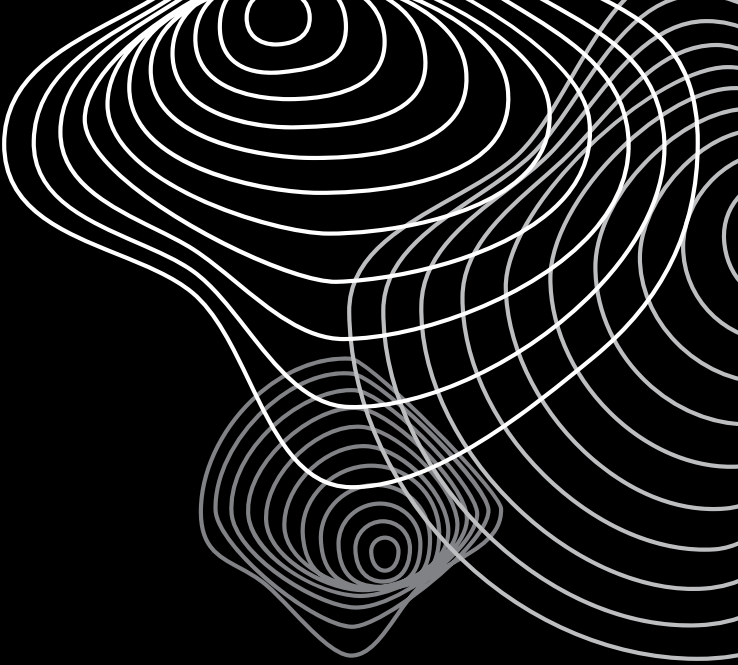




ABOUT US

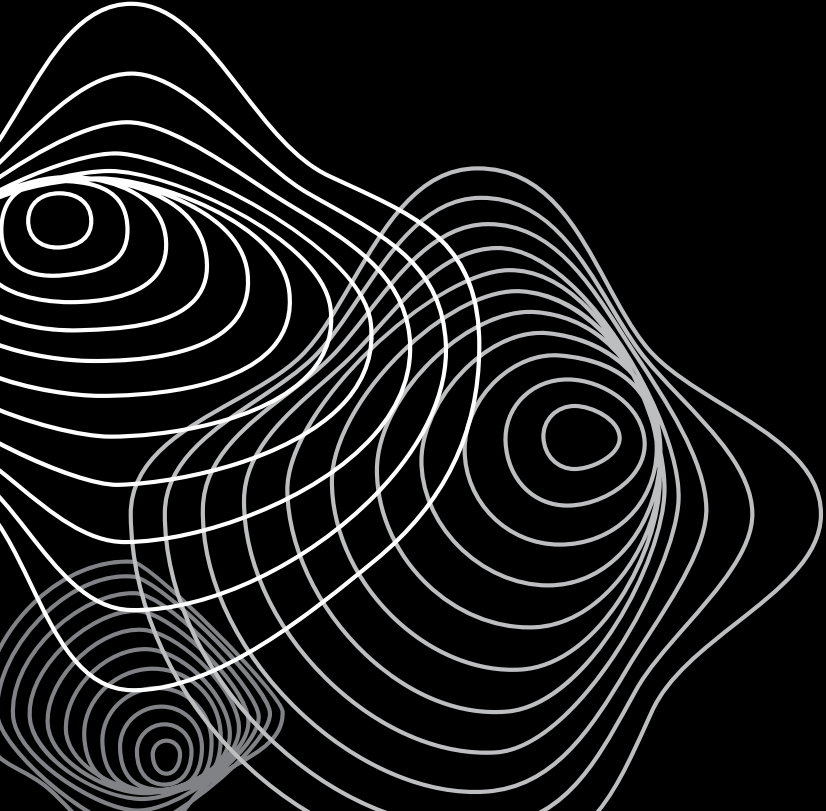
- Stateless Rest API
- Communication Interface between Unity and Server
- Industry Standard

MARIA LINDLING

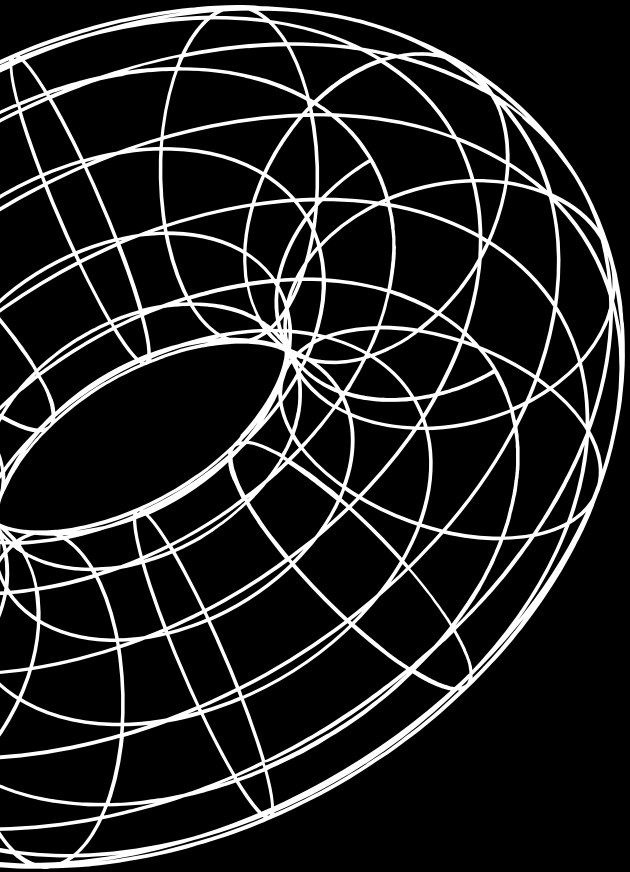


VISION

Eine robuste und skalierbare API, die dem Industriestandard gerecht wird und den Anforderungen eines Kunden in überschaubarer Zeit gerecht werden kann.



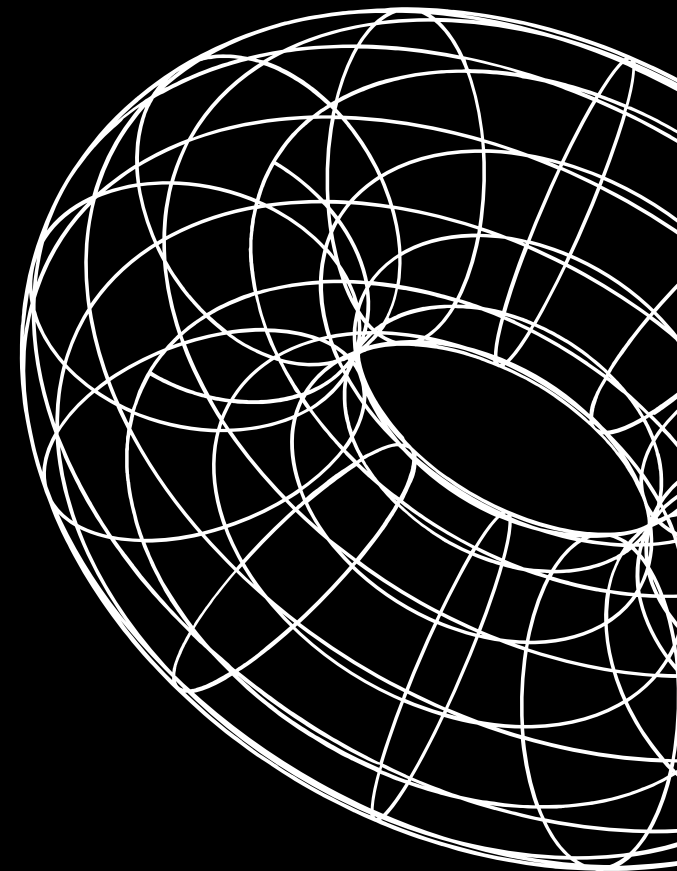
CHRIS



MISSION

Daten der User zu speichern und sie dauerhaft, sicher und anonym zur Verfügung stellen zu können, ist unsere Mission.

THIS CHRIS AGAIN



TEAM



GSAPI
Brainchild



Maria Lindling
Developer



Christof Kloninger
Developer

OVERVIEW

APP

End User Interface

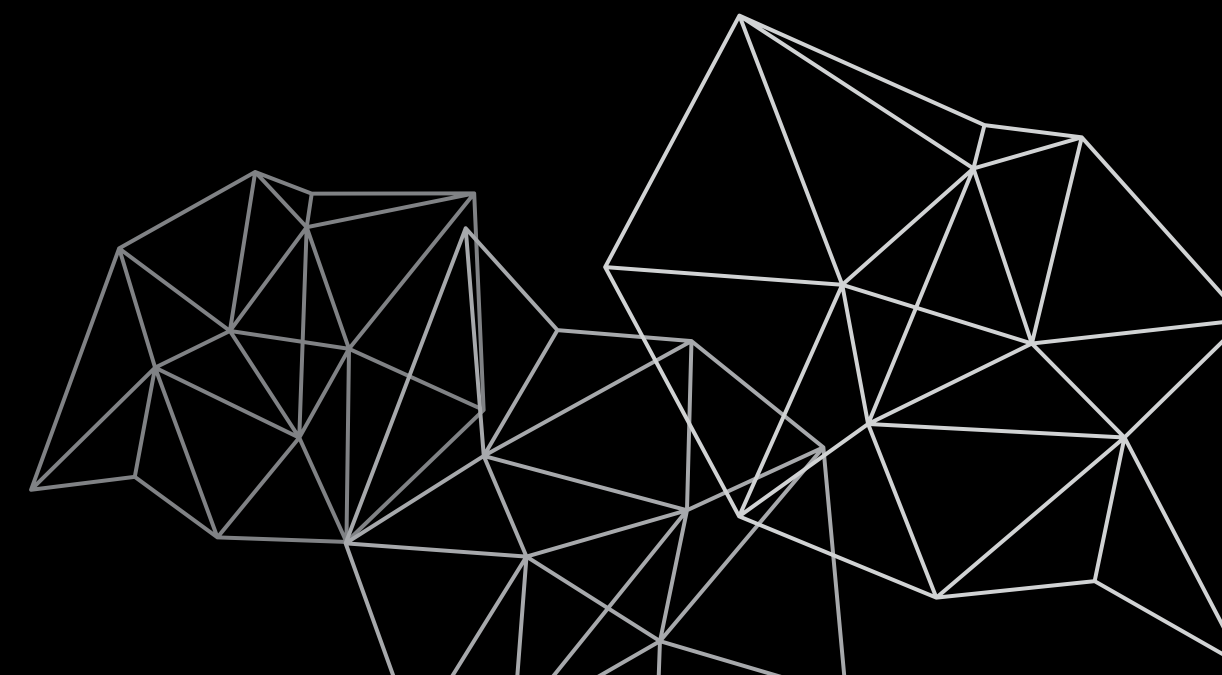
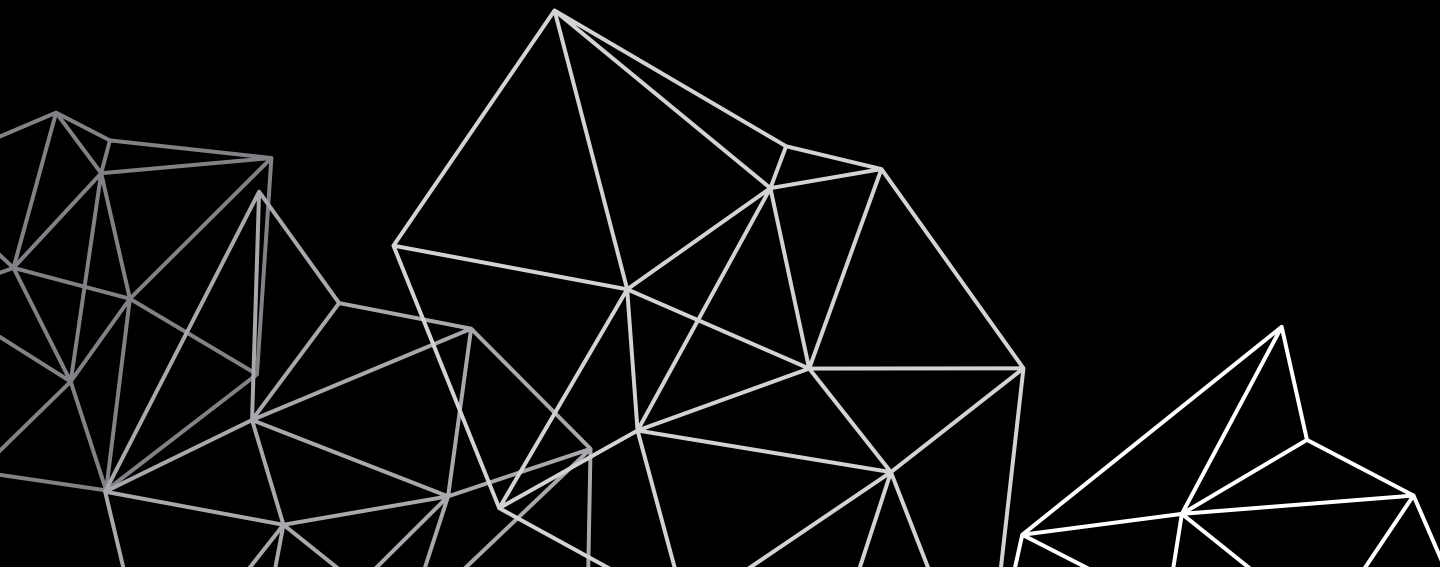
API

Application
Programming
Interface

GSAPI

Server

Databank



ROADMAP

Foundation

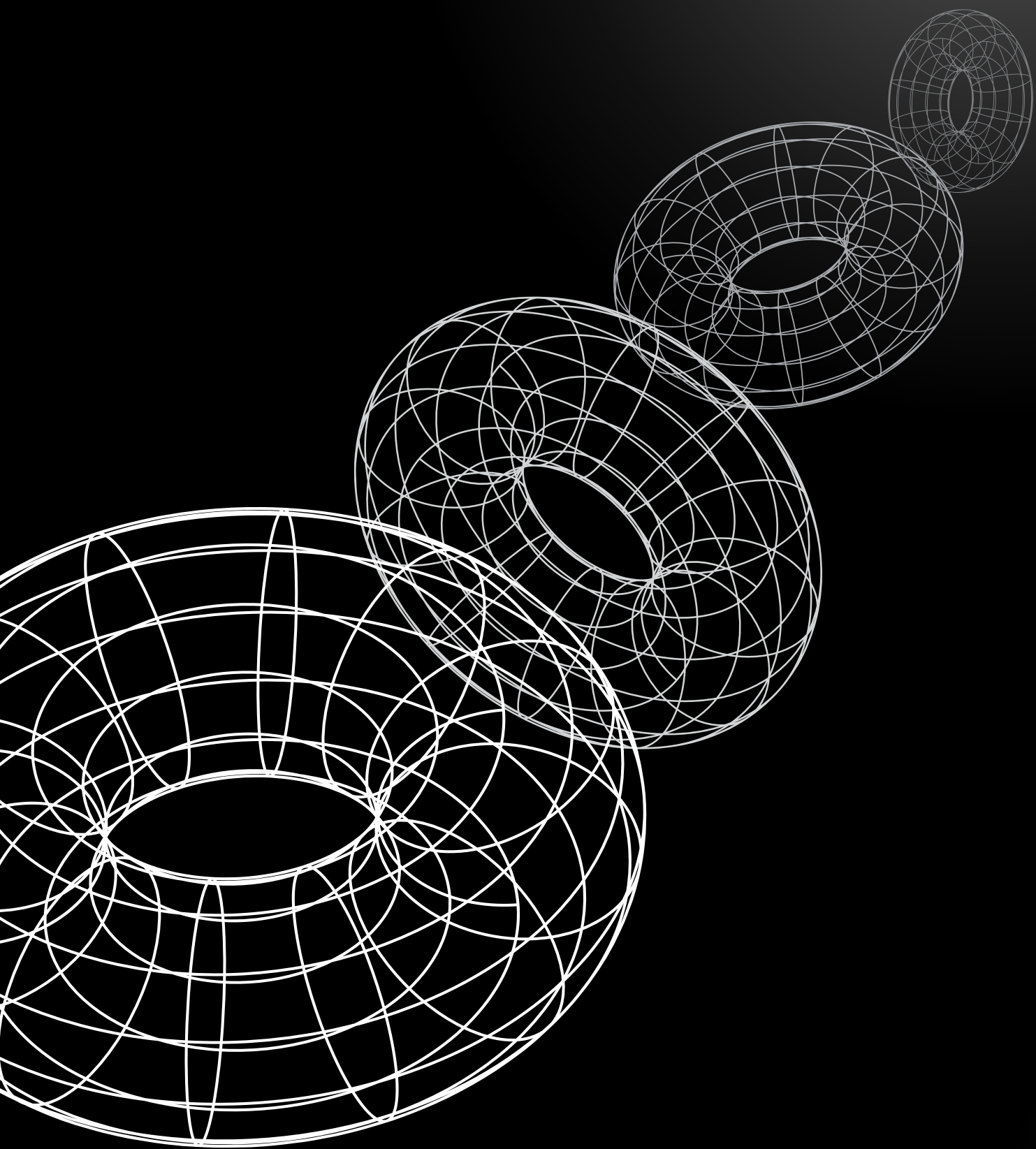
- Easy to integrate
- Secure and robust
- Performant
- Industry Standard

Advances

- Scalable
- Approachable
for Developers
- Maintainability

Future

- Deep Customization
- Expansion into more
fields
- More Data Security

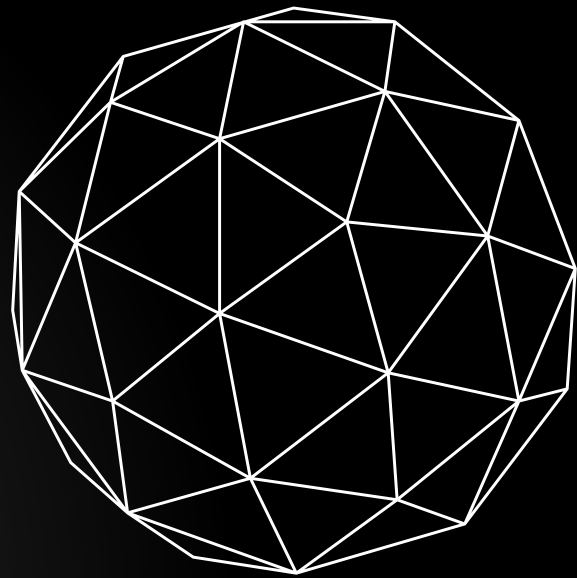


WHAT WE DO

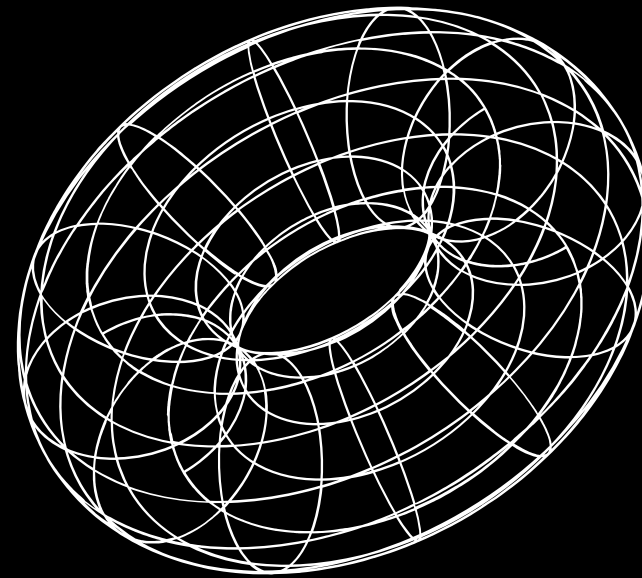
- Track
- Save
- Synchronize

GSAPI

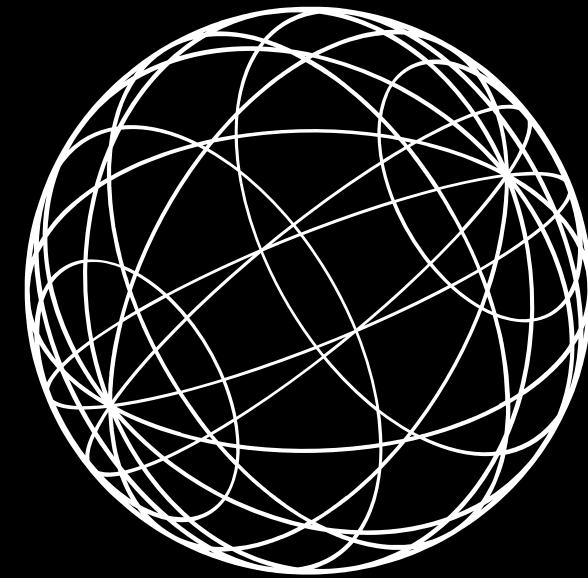
API



- Collect Data in App
- Store Data Locally



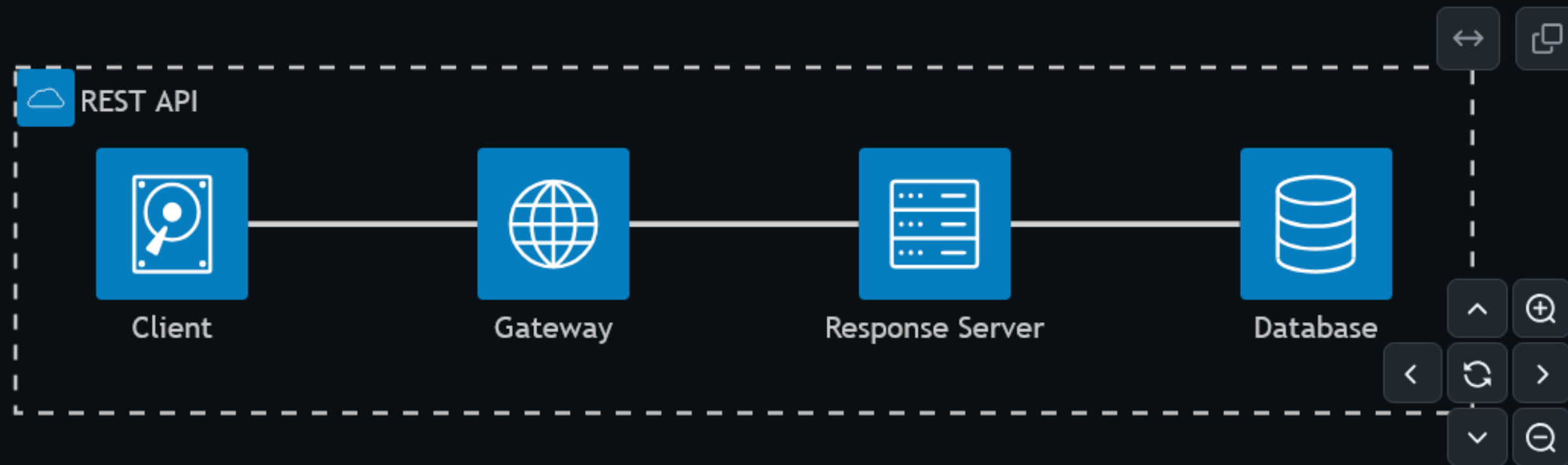
- Validate Data
- Prepare Package
- Handle Requests / Responses



- Server Connection
- CRUD

Structure

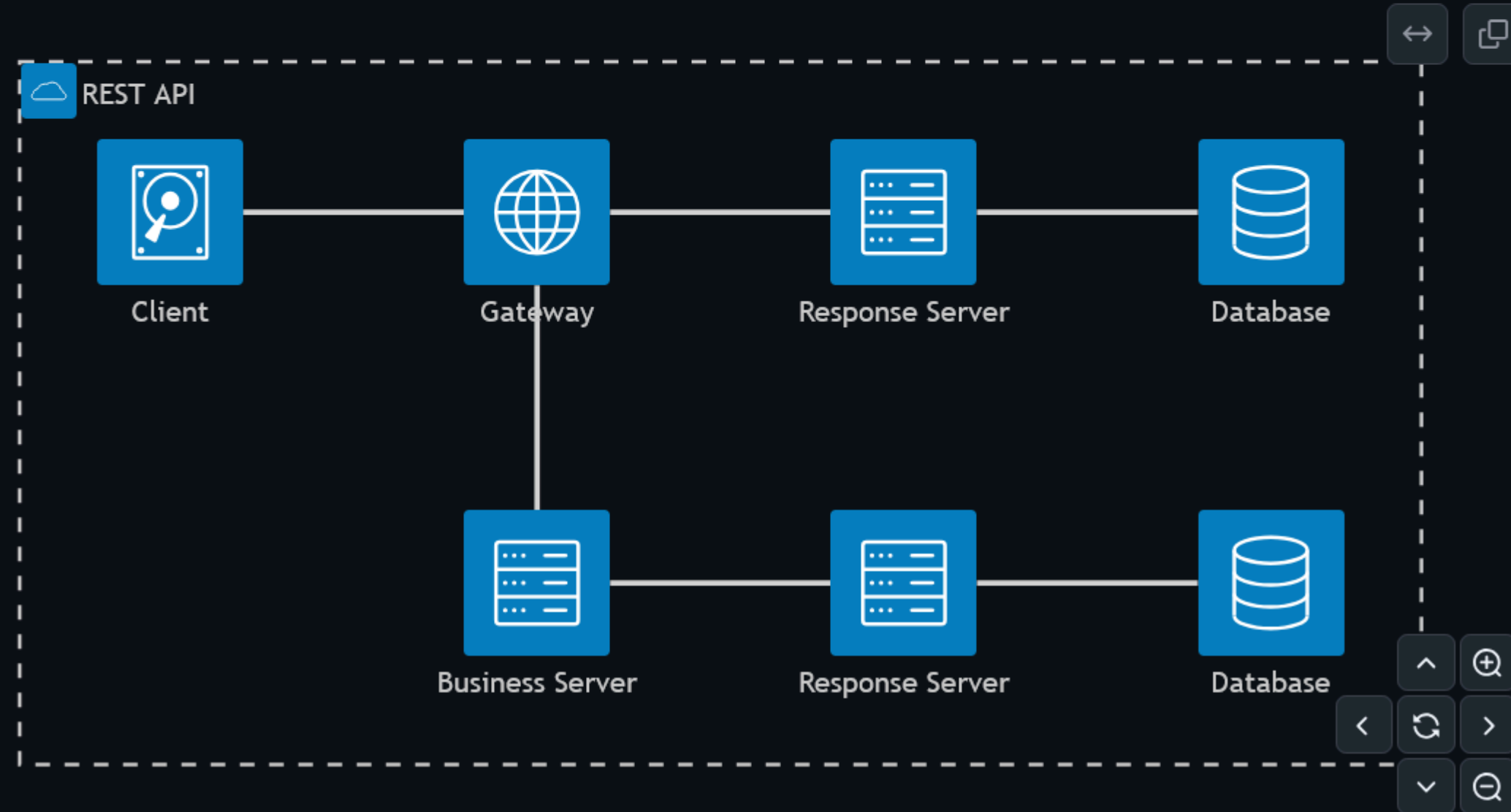
Following these principles, the api is made arbitrarily extensible, with servers able to specialize and handle specific tasks, such as security, routing, or data retrieval.



Example of a branched architecture where the API is accessed via a gateway server.



Example of a branched architecture where the API is accessed via a gateway server.



Structure

CRUD

CREATE

READ

UPDATE

DELETE

☐		Edit		Copy		Delete	7
☐		Edit		Copy		Delete	8
☐		Edit		Copy		Delete	9
☐		Edit		Copy		Delete	11
☐		Edit		Copy		Delete	53
☐		Edit		Copy		Delete	54
☐		Edit		Copy		Delete	55
☐		Edit		Copy		Delete	56

Actions

```
#region Actions: Session
```

0 references

```
> public static IEnumerable GetSessions(Action<(int,DateTime,DateTime,DateTime)>[] action, DateTime cache = default ) ...
```

0 references

```
> public IEnumerable GetSession(int sessionId, Action<int,DateTime,DateTime,DateTime> action, DateTime cache = default ) ..
```

3 references

```
public static IEnumerable AddSession(DateTime started_at, DateTime ended_at, Action<int> action)
```

```
{
```

```
    byte[] data = Encoding.UTF8.GetBytes( $"{{ \"started_at\": \"{started_at}\", \"ended_at\": \"{ended_at}\" }}" ) ;
```

```
    yield return _instance.StartCoroutine( _instance.session.Add.Call( data, You, 6 days ago • SaveManager integrat
```

```
        (text) => { action?.Invoke( JsonUtility.FromJson<PostOrPutSessionResponse>(text).insert_id ) ; }  
    ) ) ;
```

```
}
```

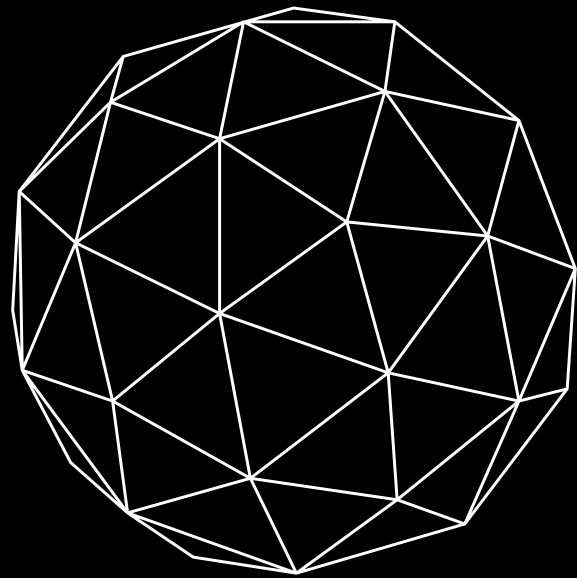
Endpoints

```
3 using System;
4 using UnityEngine;
5
6 Maria Lindling, 2 weeks ago | 1 author (Maria Lindling)
7 namespace GameStatisticsApi
8 {
9     3 references | 📦 Unity Script (3 asset references) | Maria Lindling, 2 weeks ago | 1 author (Maria Lindling)
10    internal class ApiEndpoint : MonoBehaviour
11    {
12        3 references | 📦 Unity Serialized Field
13        [SerializeField] public ApiAction GetAll ;
14        3 references | 📦 Unity Serialized Field
15        [SerializeField] public ApiAction GetOne ;
16        2 references | 📦 Unity Serialized Field
17        [SerializeField] public ApiAction Add ;
18        4 references | 📦 Unity Serialized Field
19        [SerializeField] public ApiAction Update ;
20        3 references | 📦 Unity Serialized Field
21        [SerializeField] public ApiAction Delete ;
22    }
23 }
```

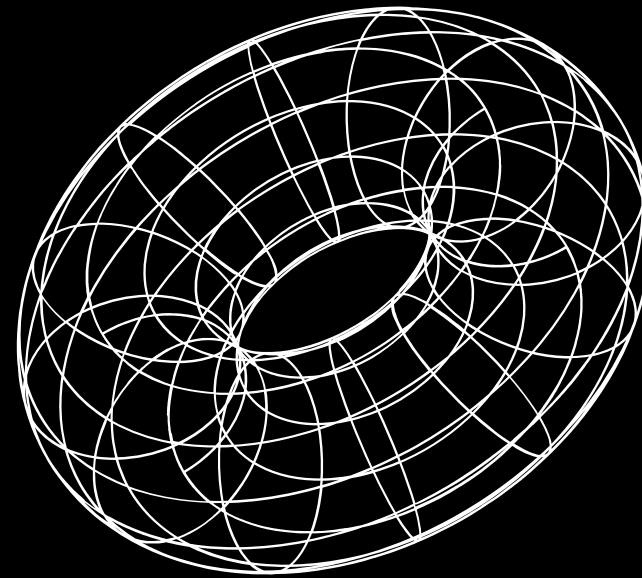
Server Connection

```
25  ✓ import { createConnection } from 'mysql2/promise';
26    import { asyncMiddleware } from './util/asyncMiddleware.js';
27    import { sessionContext } from './context/ctx_session.js';
28    import { inputContext } from './context/ctx_input.js';
29    import { timeContext } from './context/ctx_time.js';
30
31  ✓ let sql_connection = await createConnection({
32    host: "localhost",
33    user: "root",
34    password: "",
35    database: "pip_rpg",
36    namedPlaceholders: true,
37  });
38
39  ✓ sql_connection.connect(function(err) {
40    if (err) throw err;
41    console.log("Connected!");
42  });
43
```

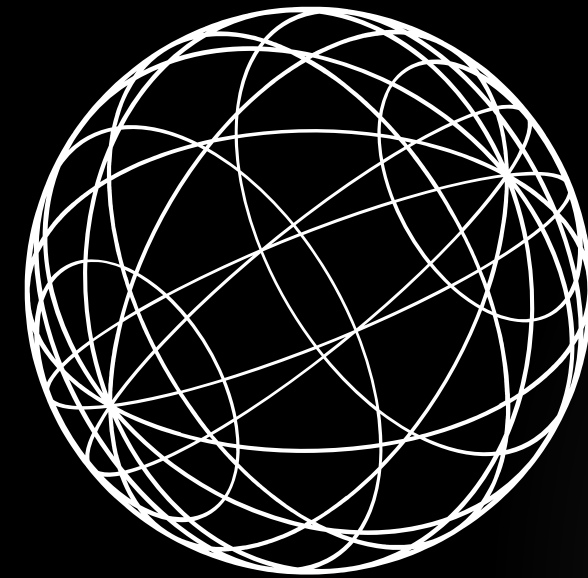
SERVER



- Build Databank
- Receive Data



- Manage Data
- CRUD



- Provide Data
- Secure Data

Databank

npc

 id <i>int(11)</i>	name <i>varchar(128)</i>	group_id <i>int(11)</i>	morale <i>int(11)</i>
1	Tutorial Agent	1	99

session

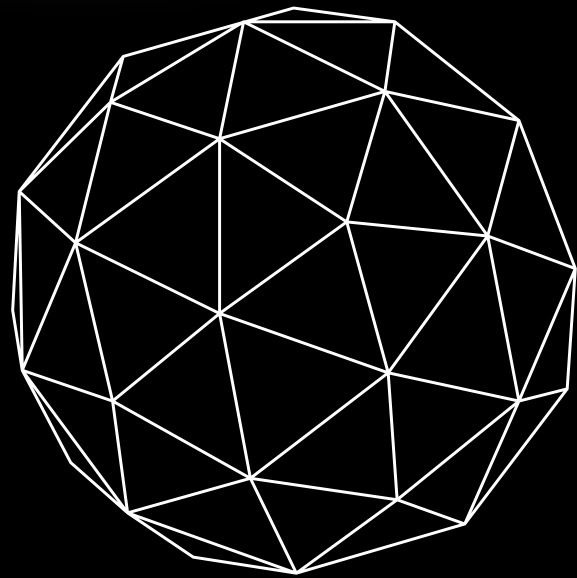
 id <i>int(11)</i>	started_at <i>date</i>	ended_at <i>date</i>	recorded_at <i>date</i>
-	-	-	-

time

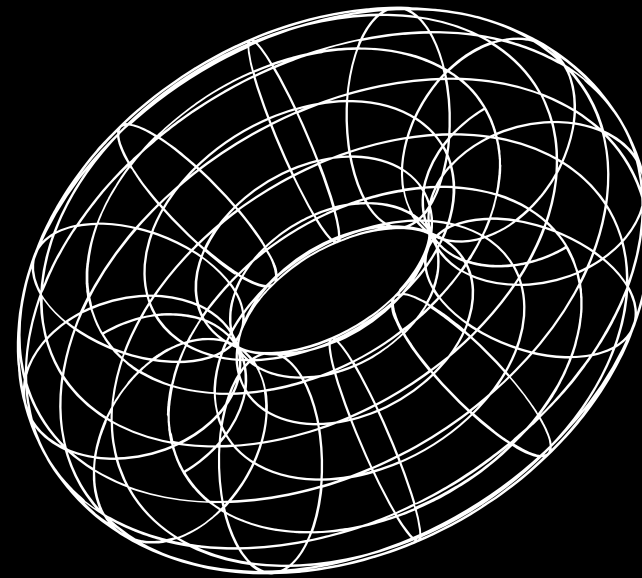
 session_id <i>int(11)</i>	in_menus <i>float</i>	in_exploration <i>float</i>	in_dialogue <i>float</i>
-	-	-	-

Copy		Delete	5	0000-00-00	0000-00-00	2026-02-09
Copy		Delete	7	0000-00-00	0000-00-00	2026-02-18
Copy		Delete	8	0000-00-00	0000-00-00	2026-02-24
Copy		Delete	9	0000-00-00	0000-00-00	2026-02-24
Copy		Delete	11	0000-00-00	0000-00-00	2026-02-24
Copy		Delete	53	0000-00-00	0000-00-00	2026-02-26
Copy		Delete	54	0000-00-00	0000-00-00	2026-02-26
Copy		Delete	55	0000-00-00	0000-00-00	2026-02-26
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Copy		Delete	57	0000-00-00	0000-00-00	2026-02-26

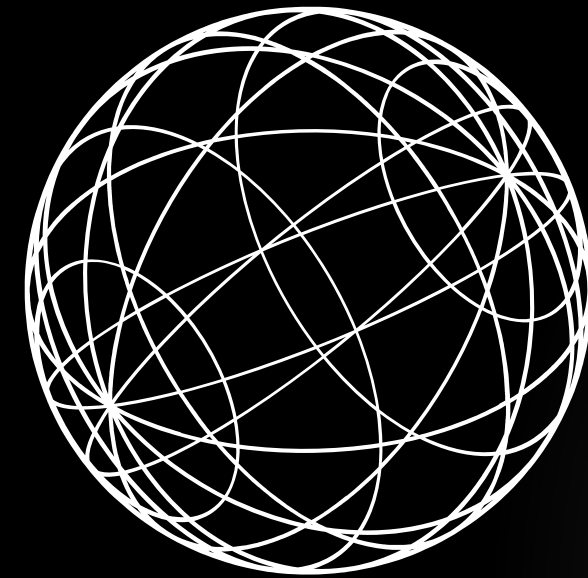
APP



- Event driven
- Track Statistics
- API Calls



- Local Data Management
- Seamless Performance



- Provide End User Interface
- Customization

Local Data Management

```
private IEnumerator SaveUserDataFile( bool isAutoSave )
{
    if( _activeSessionData == null ) yield break ;
    _activeSessionData.statistics.TimeEndedAt = DateTime.Now;
    _activeSessionData.statistics.TimeLastCache = DateTime.Now;

    string json = SerializeData(_activeSessionData);
    UserSaveData newData = DeserializeData<UserSaveData>(json);

    newData.SessionId = _activeSessionData.SessionId;
    string prefix = isAutoSave ? "AutoSave_" : "ManualSave_" ;
    string fileName = $"{prefix}{DateTime.Now:yyyyMMdd_HH:mm:ss}.dat" ;
    newData.FileName = fileName;
    newData.SlotIndex = _activeSlotIndex;

    _activeUserProfile.UserSaveDatas.Add(newData) ;

    string filesDir = Path.Combine( GetSlotPath(_activeSlotIndex) , "SaveFiles" ) ;
    if( !Directory.Exists(filesDir) ) Directory.CreateDirectory( filesDir ) ;
    byte[] encryptedData = SaveSystem.Encrypt( SerializeData( newData ) ) ;
    File.WriteAllBytes( Path.Combine( filesDir , fileName ) , encryptedData ) ;

    var existingFiles = _activeUserProfile.UserSaveDatas
        .Where(d => d.SlotIndex == _activeSlotIndex && !string.IsNullOrEmpty(d.FileName))
        .OrderBy(d => d.statistics.TimeStartedAt)
```

Interface

SaveSlot 1(ID: -1)	01/01/0001 00/00
SaveSlot 1(ID: 75)	26/02/2026 18/10
SaveSlot 1(ID: 75)	26/02/2026 18/10
SaveSlot 1(ID: 76)	26/02/2026 19/07
SaveSlot 1(ID: -1)	01/01/0001 00/00
SaveSlot 1(ID: -1)	01/01/0001 00/00
SaveSlot 1(ID: 77)	26/02/2026 21/07
SaveSlot 1(ID: 77)	26/02/2026 21/07

SaveSlot 2(ID: 74)	26/02/2026 16/18
SaveSlot 2(ID: 74)	26/02/2026 16/18
SaveSlot 2(ID: 74)	26/02/2026 16/18
SaveSlot 2(ID: 74)	26/02/2026 16/18
SaveSlot 2(ID: 76)	26/02/2026 19/07

SaveSlot 3(ID: 74)	26/02/2026 16/18
SaveSlot 3(ID: 74)	26/02/2026 16/18
SaveSlot 3(ID: 74)	26/02/2026 16/18
SaveSlot 3(ID: 77)	26/02/2026 21/07
SaveSlot 3(ID: 77)	26/02/2026 21/07
SaveSlot 3(ID: 77)	26/02/2026 21/07

Slots

Save

Delete

Load

Back

SaveSlot 1(ID: -1)	01/01/0001 00/00
SaveSlot 1(ID: 75)	26/02/2026 18/10
SaveSlot 1(ID: 75)	26/02/2026 18/10
SaveSlot 1(ID: 76)	26/02/2026 19/07
SaveSlot 1(ID: -1)	01/01/0001 00/00
SaveSlot 1(ID: -1)	01/01/0001 00/00
SaveSlot 1(ID: 77)	26/02/2026 21/07
SaveSlot 1(ID: 77)	26/02/2026 21/07
SaveSlot 1(ID: 77)	26/02/2026 21/07
SaveSlot 1(ID: 77)	26/02/2026 21/07
SaveSlot 1(ID: -1)	01/01/0001 00/00

SaveSlot 2(ID: 77)	26/02/2026 21/07
--------------------	------------------

SaveSlot 3(ID: 74)	26/02/2026 16/18
SaveSlot 3(ID: 74)	26/02/2026 16/18
SaveSlot 3(ID: 74)	26/02/2026 16/18
SaveSlot 3(ID: 77)	26/02/2026 21/07
SaveSlot 3(ID: 77)	26/02/2026 21/07
SaveSlot 3(ID: 77)	26/02/2026 21/07

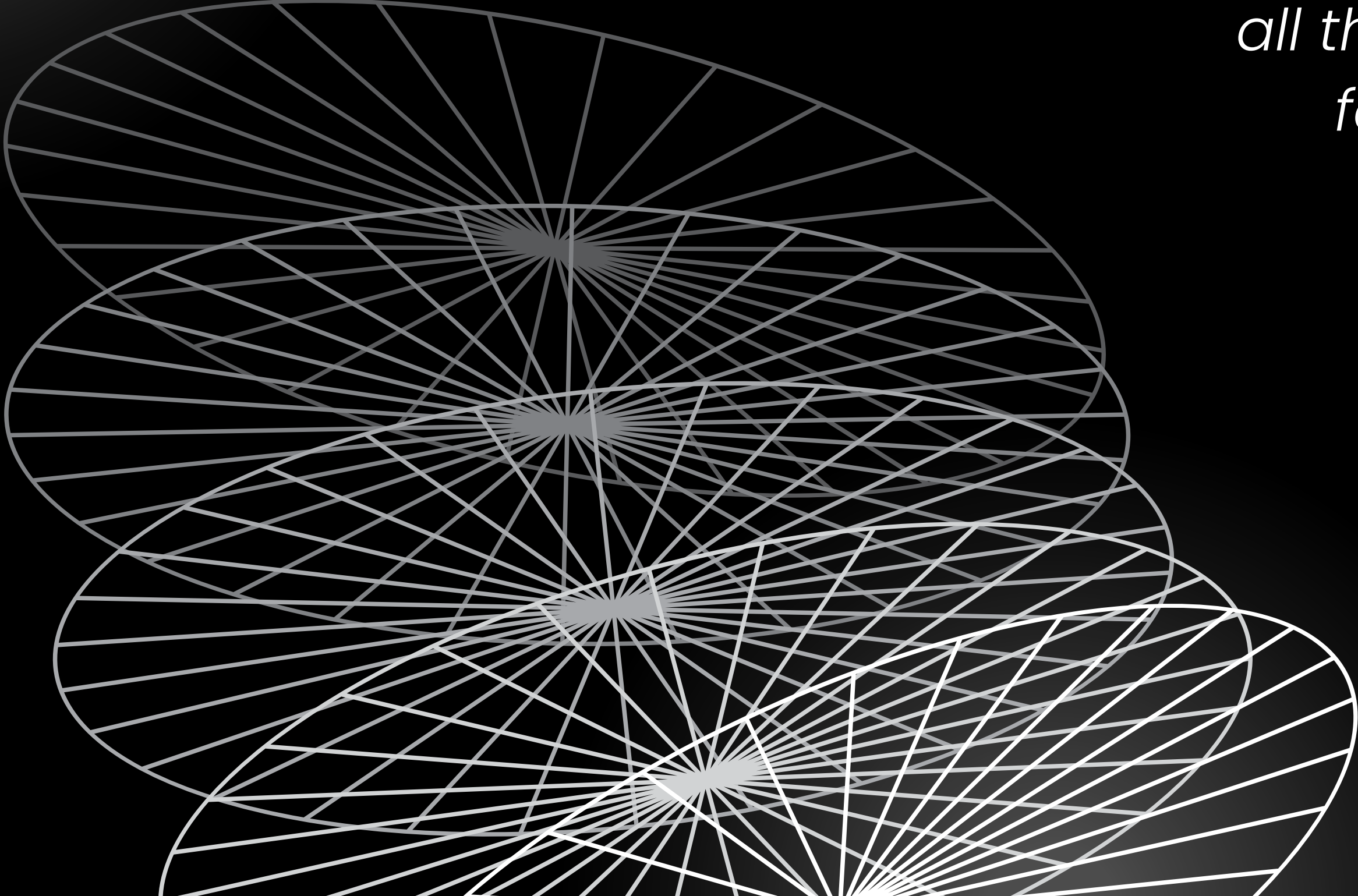
Slots

Save

Delete

Load

Back



*Humanity is acquiring
all the right technology
for all the wrong
reasons.*

R. BUCKMINSTER FULLER



THANK YOU

GSAPI

Quellen

Green API png – [Qaisir Mehmood](#)

GS API – [Link to our project](#)

GSAPI [Project Management](#) – [Link](#)